Updated:OCT 22/2025

													1/4" HRS			
					14ga COPPER .084						3/16" CRS .187					
				16ga COPPER .063					10ga CRS .135							
			18ga COPPER .048					11ga CRS 120								
		24ga COPPER .020					12ga CRS .104									
					14ga CRS .074											
					16ga CRS .060						1/4" AL.250					
				18ga CRS .048					3/16 AL.188							
			20ga CRS .036					1/8" AL .125								
		22ga CRS .030				10ga AL.102										
		24ga CRS .025			11ga AL.090								10ga SS .135			
					12ga AL.081							11ga SS .125				
				14ga AL.064					12ga SS .108							
				16ga AL.051				14ga SS .078								
			18ga AL.040				16ga SS .062									
		20ga AL.032				18ga SS .050							1/4" COPPER			
		22ga AL.025			20ga SS .037						3/16" COPPER					
		24ga AL.020		22ga SS .030					1/8" COPPER							
PUNCH TYPE		0.003	0.004	0.006	0.008	0.010	0.012	0.016	0.020	0.024	0.027	0.030	0.038	0.040	0.050	WHSP
Single Round																
0.063					1											
0.094	6-32 PP			1				1								
0.125				1			1		1							
0.130					1		1									
0.142							1		1							
0.156				1			1		1		1					
0.166											1					
0.177				1			1		1							
0.191					1		1	1			1					
0.196				1			1		1							
0.201									1							
0.203				1			1									
0.213							1		1		1					
0.221							1		1							
0.228				1			1		1							
0.236				1			1				1			1		
0.250				1			1		1		1			1	1	
0.266							1		1							
0.272							1		1							
0.281				1	1		1		1			1				

0.313									1							
0.323									1							
0.328				1			1		1							
0.344									1							
0.375							1				1	0.034				
0.397				1					1							
0.406									1							
0.416								1			1					
0.438	3/5			1			1	1							1	
0.450			1				1		1							
0.472						1	1				1					
0.500	4/5			1			1		1	1		1		1		
0.530				1			1		1		1					
0.563				1			1	1	1				1			
0.625				1				1			1			1		
0.650					1											
0.656							1		1		1	0.034				
0.688				1				1								
0.711						1										
0.750				1				1			1			1		
0.813				1				1								
0.875				1				1	1		1			1		
0.910							1				1			1		
0.938				1			1	1								
1.000				1				1	1		1		1			
1.063								1	1							
1.125				1				1								
1.250				1				1								
1.375				1				1								
1.500				1				1								
1.625				1				1								
1.750				1				1								
2.000				1				1								
2.250							1		1							

2.500							1		1							
3.000				1				1								
PUNCH TYPE		0.003	0.004	0.006	0.008	0.010	0.012	0.016	0.020	0.024	0.027	0.030	0.038	0.040	0.050	WHSP
Multi 10 Round	Maximum Punching Material thickness .177"															
0.063				1	1			1								
0.076				1		1		1								
0.094				1	1		1	1								
0.111							1		1							
0.125	10/11				1		1	1		1						
0.136								0.017	0.018							
0.156	9/11			1			1	1		1	1					
0.166	8/12			1												
0.177							0.013	1								
0.188	1/12			1	1		1	1		1						
0.191					1			1		1						
0.196	4/11						1			1						
0.203	1/11			1			1	1	1	1						
0.205							1					1				
0.213	6/11			1			1	1	1	1						
0.219								1								
0.221	6/12															
0.226						0.009										
0.228	7/12			0.007			1		1							
0.250	5/11			1			1	1		1						
0.266							1									
0.272	10/12			1			1				1					
0.281	7/11			1			1		1		1					
0.290				1					0.021		1					
0.313	2/12				1		1	0.018			1					
0.328						1	1									
0.344	8/11						1	1			1					
0.360				1				1								
0.375	3/11			1			1		1	1	1					
0.406	3/12			1			1	1			1					

PUNCH TYPE		0.003	0.004	0.006	0.008	0.010	0.012	0.016	0.020	0.024	0.027	0.030	0.038	0.040	0.050	WHSP
Single Square																
0.125							1		1	1						
0.200				1			1		1		1					
0.250				1			1		1		1			1		
0.325							1		1		1					
0.310							1		1		1					
0.318							1		1		1					
0.325							1		1		1					
0.380	2/5					1	1		1		1		1			
0.400							1		1		1					
0.500				1			1		1	1				1		
0.750			1					1					1			
1.000				1				1				1				
1.250							1			1						
2.000				1				1								
Multi 10 square	Maximum punching material thickness .177"															
0.250	4/12			1			1	1	1							
PUNCH TYPE		0.003	0.004	0.006	0.008	0.010	0.012	0.016	0.020	0.024	0.027	0.030	0.038	0.040	0.050	WHSP
Single Rectangle																
.062x0.250						1	1		1							
.078x.750							1		1							
.126x.251					1		1		1		1					
.125x.750							1		1							
.135x1.050				1			1		1							
.187x.312					1		1	1	1		1					
.394x.787				1		1		1				1		1		
.109x.506	1/5						1		1		1					
.141x.506							1				1					
.250x1.000				1			1		1		1			1	1	

.250x1.500			1		1											
.200x2.000				1	1		1		1	1	1			1		
.125x2.000				1			1		1			1				
.125x2.900			1	1			1		1	1	1					
Multi 10 Rectangle	Maximum punching material thickness .177"															
.126x.251	5/12			1			1	1	1	1	1					
.187x.312	2/11			1	1		1	1	1							
.215x.250				1			1	1								
MultiShear																
.197x3.000																
PUNCH TYPE		0.003	0.004	0.006	0.008	0.010	0.012	0.016	0.020	0.024	0.027	0.030	0.038	0.040	0.050	WHSP
Single Oblong																
.170x1.500								1			1					
.187x0.500			1				1		1							
.250x1.500							1	1	1		1					
.275x.984					1						1					
.281x.450	5/5						1		1							
.313x.480							1		1							
.313x.625							1		1		1					
.406x.550							1		1							
.438x.600							1		1							
.510x1.260											1					
.150x.280							1		1							
.250X2.000						1	1		1							
.188x2.600						1										
Multi Oblong	Maximum punching material thickness .177"															
.100x.400				1				1								
.170x.375				1				1								
.187x.312					1	1	1		1							
.250x.307				1				1								

.327x.406				1				1								
PUNCH TYPE		0.003	0.004	0.006	0.008	0.010	0.012	0.016	0.020	0.024	0.027	0.030	0.038	0.040	0.050	WHSP
Double D																
.660x.750	GPT101				1											
Key way punch																
.625x.375x.190	GPT201			1												
Special punch																
	GPT202										1					
	GPT203										1					
Line Stamp	GPT204										1					
R5.0 Radius	GPT206								0.021"							
Extrusion Tools																
4-40	GPT301															
6-32	GPT302															
Tapping Tools																
4-40	GPT401															
6-32	GPT402															
Single Louver Tool																
3.65x2.73	GPT501															
single Louver	GPT502	12X5X60 Upward louver														
Multi-Bend																
Small 0.787	GPT801	1263999-00														52
Large 2.165	GPT802	0688738-05														52
Radius Tools																
R0.25	GPT901				1			1		1						
K.O. Tools																
KO.875 DN1ST	GPT701															
KO1.109 UP2ND	GPT702															

KO1.375 DN1ST	GPT703															
KO1.734 UP2ND	GPT704															
.903 Dimple	GPT705															
Roller Tools																
Roller Offset	GPT601	Tool type: RollerFold 2x21 Ks14130756														
Roller Beading	GPT602	Use Radius 0.8" for corners and must show at center,Tool tpe: RollerBead 2T Ks14130755														

Formi 0.27x0.27 for superior glass SPG-01-CV-SS, C'sink P.P.=Ø.266 OD=.320" ID=.265"

Center punch .0394

Manual C'sink 82° size .250, .375, .500, .750, 1", Punch center punch for C'sink P.P.

Louver tool minimum distance from center to center 2.6" (side by side) x 0.9" (bottom to bottom)

Top punch for forming radius .250 for most length

Top punch for forming radius .750 max length is 16"

Hand hole cover use 14ga CRS

6-32 Tapping PP for .125AL is Ø.094

Copper Bar .250 use .020 die

Programmer note: Do not use dyn. MT. ID 01999011 tools are set for no change. ID 01999012 tool can be changed from 7/10 to 10/10. Only 11 tools could be setup in every program.

FR4--Material thickness is .125" Sheet size is 36"x48"

FISH PAPER--Material thickness is .030" Sheet size is 40"x27"